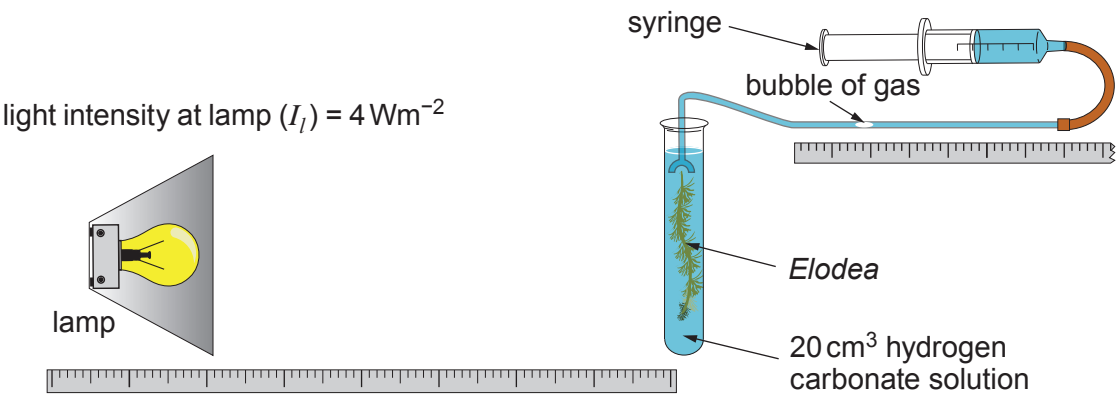


3. A group of students used the apparatus shown in **Image 3.1** to investigate the effect of light intensity on the rate of photosynthesis. The volume of gas was measured after a minute by drawing it up with the syringe and measured using the scale.

**Image 3.1**



The students' results are shown in **Table 3.2**:

**Table 3.2**

| Distance from lamp / m | Distance from lamp <sup>2</sup> / m <sup>2</sup> | Light intensity at plant / Wm <sup>-2</sup> | Volume of gas / mm <sup>3</sup> min <sup>-1</sup> |         |         |      |
|------------------------|--------------------------------------------------|---------------------------------------------|---------------------------------------------------|---------|---------|------|
|                        |                                                  |                                             | Trial 1                                           | Trial 2 | Trial 3 | mean |
| 0.30                   | 0.0900                                           | 44                                          | 1222                                              | 978     | 1098    | 1099 |
| 0.25                   | 0.0625                                           | 64                                          | 2098                                              | 1897    | 2134    | 2043 |
| 0.20                   | 0.0400                                           | 100                                         | 3302                                              | 2788    | 3100    | 3063 |
| 0.15                   | 0.0225                                           | 178                                         | 4012                                              | 3876    | 4100    | 3996 |
| 0.10                   | .....                                            | .....                                       | 3998                                              | 3887    | 4100    | 3995 |
| 0.05                   | 0.0025                                           | 1600                                        | 4011                                              | 3888    | 3998    | 3966 |

- (a) **Complete Table 3.2** using information from **Image 3.1** and the formula for light intensity below: [3]

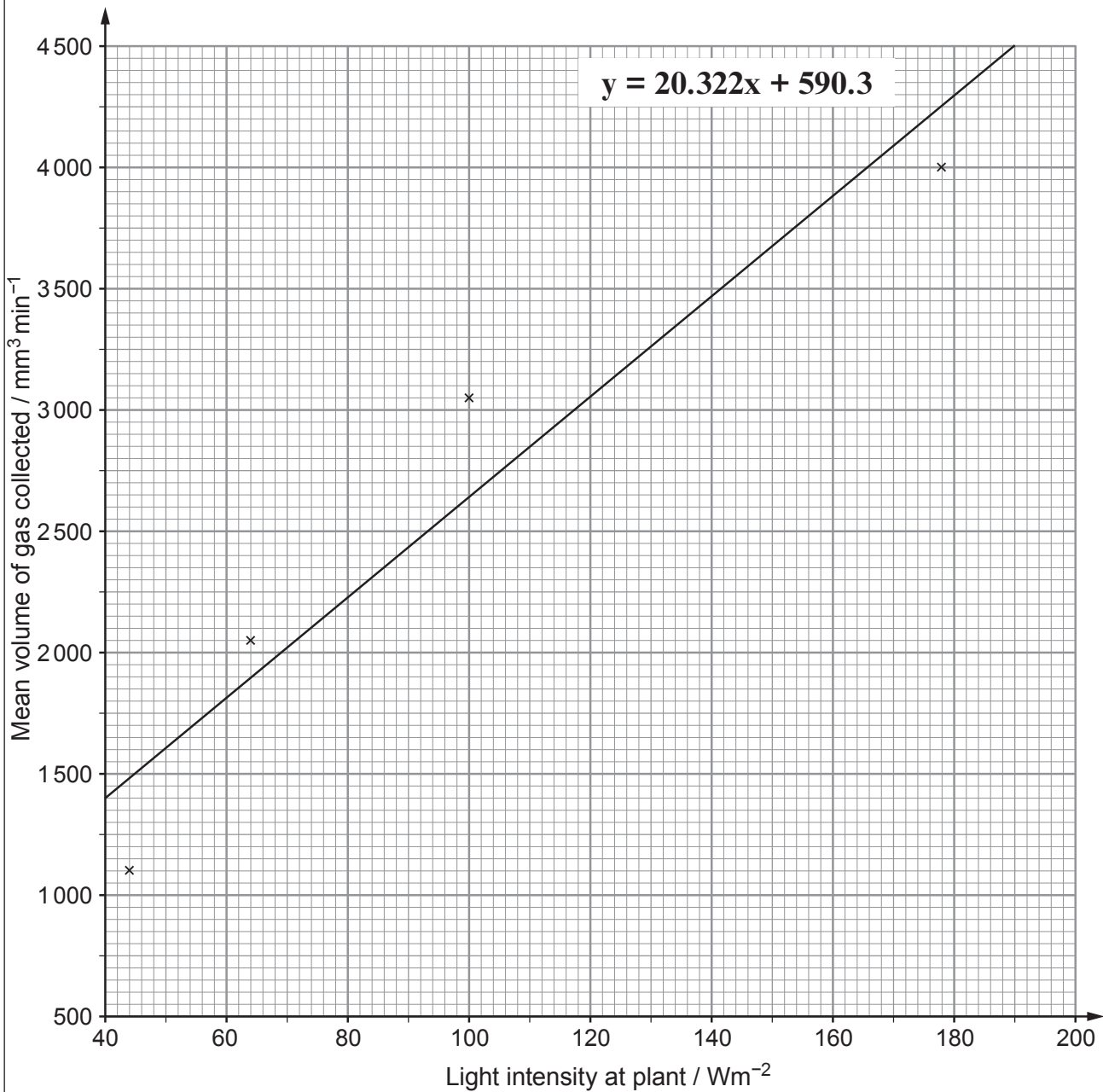
Light intensity at plant,  $I_p = I_l \left( \frac{1}{d^2} \right)$

Where:  $I_p$  = light intensity at plant,  
 $I_l$  = light intensity at lamp,  
 $d$  = distance from lamp in metres.



Some of the results were plotted in **Graph 3.3**. The students assumed that the mean volume of gas collected was directly proportional to light intensity. The equation for their line of best fit is shown above the line.

**Graph 3.3**



- (b) (i) Use the equation given on **Graph 3.3** and the relevant values from **Table 3.2** to calculate the mean volume of gas that should have been collected when the lamp was 0.15 m from the plant. [2]

Expected mean volume of gas = ..... mm<sup>3</sup> min<sup>-1</sup>



Examiner  
only

(ii) State **one** environmental factor that should have been maintained at a constant value during the experiment. [1]

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(iii) The **expected** mean volume of gas collected (calculated from the equation on **Graph 3.3**), when the lamp was 0.05 m from the plant, was  $33\,105.5\text{ mm}^3\text{ min}^{-1}$ . Explain why the expected volume of gas was higher than the volume **actually** collected when the lamp was 0.05 m from the plant (shown in **Table 3.2**). [2]

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